

Gaurav Ghop

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Backend & integration engineer with 3 years building connector, ingestion, and sync-reliability systems that normalize messy, heterogeneous data across PLM, ERP, CRM, MES and other enterprise systems. **U.S. work authorized (EAD) — no sponsorship required.**

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, C

Backend & Integration: Connectors, iPaaS, Webhooks, OAuth, Schema Mapping, Data Transformation, Data Normalization, Bi-directional Sync, Reconciliation, Idempotency, DLQ, Event-Driven Architecture, REST / OpenAPI / OData, Microservices, Spring Boot, FastAPI

Distributed Data: Kafka, PostgreSQL, PostGIS, Elasticsearch, Redis, WAL / SSTables (LSM)

Cloud & Infra: AWS / Amazon Web Services (Bedrock, SQS), Docker, Kubernetes, CI/CD, CloudWatch, Linux

AI & Agents: LLM Integration, Tool-use & MCP, Multi-step Planning, Guardrails & Evaluation, Local/Edge Inference

EXPERIENCE

Tiny Archives

Jun 2025 – Aug 2025

Software Engineer, Summer Intern

New York, NY

- Refactored core service modules into a shared, reusable scaffolding template so the team could stand up new endpoints from a common contract instead of re-writing boilerplate, **cutting new-endpoint setup from 6 hours to 90 minutes.**
- Built a format-agnostic parsing service that unified heterogeneous archival exports (**EAD/XML, DOCX, CSV, PDF**) into one normalized model behind a single **API**, so the app could search and export every source uniformly regardless of its original structure.

Duende Ingenious

Jul 2023 – Aug 2024

Founding Software Engineer

Pune, India

- Designed a rider-aggregator **API** that abstracted multiple delivery providers behind one interface, backed by an **SQS** queue with retries and a **dead-letter queue** so a provider timeout parked the order for retry instead of dropping it silently, **cutting failed orders 40%.**
- Decomposed a monolithic order service into four **event-driven microservices** with idempotent handlers and **circuit breakers**, isolating failures so a slow downstream provider degraded one path instead of stalling the pipeline, **cutting cycle time from 10 days to 6.**

EQ Technologic

Aug 2021 – Jun 2023

Software Engineer I

Pune, India

- Cut incident triage time **18%** with correlation IDs and structured per-tenant logging that let on-call engineers trace a failure to the exact tenant and connector across defense deployments (incl. **Lockheed Martin, US Navy**) instead of searching blind.
- Owned sync reliability for a **PLM/ERP** platform, building **Java Spring Boot** and **Kafka** consumers with idempotent processing, dead-letter routing, and automated reconciliation so a change in **Teamcenter/Windchill** never left **SAP** showing a stale part revision, across multi-tenant deployments.
- Exposed unified part and BOM data as **REST and OData** services over the platform's **Common Data Model**, giving downstream apps one semantically consistent API instead of brittle point-to-point integrations against each PLM and ERP object model.
- Automated pre-deploy validation across the platform's **100+ connectors** with per-connector smoke tests, drift detection, and one-command rollback, **replacing a 2-hour manual checklist** and catching breakages before hybrid cloud and on-prem rollouts.

EDUCATION

New York University

Sept 2024 – May 2026

M.S. Computer Engineering

New York, NY

Relevant coursework: Distributed Systems, High-Performance Machine Learning, Design & Analysis of Algorithms.

Vishwakarma Institute of Information Technology

Aug 2017 – May 2021

B.Tech. Electronics & Telecommunications Engineering

Pune, India

PROJECTS

Shadow Engineer, Onboarding Copilot | *AWS Bedrock, MCP, Tool-use Agents, RAG* | [LINK](#)

- Built an onboarding agent on AWS Bedrock that assembles the context for a ticket using MCP tools across Jira, the repo, and commit history, then cites the exact files and commits it used so a new engineer can verify each suggestion.

TinyLSM KV Store | *C, WAL, SSTables* | [LINK](#)

- Implemented an LSM-tree key-value store in C with a write-ahead log for crash recovery and leveled compaction to bound read amplification, plus Bloom filters so point lookups skip SSTables that cannot hold the key, sustaining 80K ops/sec at p95 2.5 ms on 20 GB.

ACHIEVEMENTS

Merit Scholarship for graduate tuition, NYU • **Runner-Up** AWS GenAI Hackathon 2025 • **Finalist** Google DeepMind Hackathon 2025 • **Runner-Up** Smart India Hackathon 2020 • Google KickStart Round B 2020 • ACM ICPC Regionals 2019 • CodeChef 4★